

Homework Assignment 3

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1. (1.B.5) Give an example of a sequence of continuous real-valued functions $f_1, f_2, \dots$ on $[0, 1]$ and a continuous real-valued function f on $[0, 1]$ such that

$$f(x) = \lim_{k \rightarrow \infty} f_k(x)$$

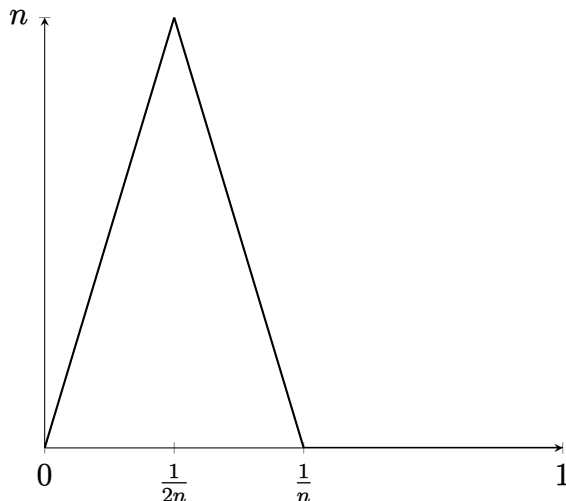
for each $x \in [0, 1]$ but

$$\int_0^1 f \neq \lim_{k \rightarrow \infty} \int_0^1 f_k.$$

Proof. Define

$$f_n(x) = \begin{cases} 2n^2x & 0 \leq x < \frac{1}{2n} \\ -2n^2x + 2n & \frac{1}{2n} \leq x < \frac{1}{n} \\ 0 & \frac{1}{n} \leq x \leq 1, \end{cases}$$

so that $\int_0^1 f_n = \frac{1}{2}$ for all $n \in \mathbb{Z}^+$ as shown below.



But the limit f_n as $n \rightarrow \infty$ is $f(x) = 0$, since $f_n(0) = f_n(1) = 0$ for all n , and for any $x \in (0, 1)$ there is an n large enough that $\frac{1}{n} < x$, so $f_m(x) = 0$ for all $m \geq n$. Since $\int_0^1 f = 0$ and $\lim_{n \rightarrow \infty} \int_0^1 f_n = \frac{1}{2}$, we have our contradiction. $\square$

2. (2.A.9) Prove that $|A| = \lim_{t \rightarrow \infty} |A \cap (-t, t)|$ for all $A \subset \mathbb{R}$.

Proof. Using a result from the previous homework, we have

$$|A| = |A \cap (-t, t)| + |A \cap (\mathbb{R} \setminus (-t, t))| \geq |A \cap (-t, t)| \quad (1)$$

for any $t > 0$. By monotonicity, $|A \cap (-t, t)| \leq |A \cap (-(t + \varepsilon), t + \varepsilon)|$ for any $\varepsilon > 0$, so we can take the supremum over all such t in (1) to get

$$|A| \geq \sup_{t>0} |A \cap (-t, t)| = \lim_{t \rightarrow \infty} |A \cap (-t, t)|.$$

For the inequality in the other direction, we can break $|A|$ up like so:

$$\begin{aligned} |A| &= \left| \bigcup_{n=-\infty}^{\infty} (A \cap [n-1, n]) \right| \\ &\leq \sum_{n=-\infty}^{\infty} |A \cap [n-1, n]| \\ &= \lim_{N \rightarrow \infty} \sum_{n=-N}^N |A \cap [n-1, n]| \\ &= \lim_{N \rightarrow \infty} \sum_{n=-N}^N |A \cap (n-1, n]| \\ &= \lim_{n \rightarrow \infty} |A \cap (-n, n]| \\ &= \lim_{n \rightarrow \infty} |A \cap (-n, n)| \\ &= \lim_{t \rightarrow \infty} |A \cap (-t, t)|. \end{aligned}$$

□

3. (2.A.11) Prove that if $I_1, I_2, \dots$ is a disjoint sequence of open intervals, then

$$\left| \bigcup_{k=1}^{\infty} I_k \right| = \sum_{k=1}^{\infty} \ell(I_k).$$

Proof. This follows immediately upon noticing two key relationships. First, by countable subadditivity,

$$\left| \bigcup_{k=1}^{\infty} I_k \right| \leq \sum_{k=1}^{\infty} |I_k| = \sum_{k=1}^{\infty} \ell(I_k).$$

Second, notice that the collection of intervals $I = \{I_k\}_{k \in \mathbb{Z}^+}$ is itself an open cover of $\bigcup_{k=1}^{\infty} I_k$. For any open cover $J = \{J_k\}_{k \in \mathbb{Z}^+}$ of $\bigcup_{k=1}^{\infty} I_k$ consisting of open intervals, define a collection $K = \{J_k \cap (\bigcup_{j \in \mathbb{Z}^+} I_j)\}$, then reindex so that we have a collection of open intervals. This collection still covers I but has the property that for each k , $I_k \subset \bigcup \{J_j\}$, where each such union is disjoint. We can now compare I to K and K to J like so:

$$\left| \sum |I_k| \right| \leq \left| \sum |K_k| \right| \leq \left| \sum |J_k| \right|.$$

Thus $|\sum I_k|$ is a lower bound on $\{\sum_{k=1}^{\infty} \ell(L_k) : \bigcup I \subset \bigcup L_k\}$, and so $|\sum I_k| = \sum_{k=1}^{\infty} \ell(I_k)$. □

4. (2.B.5) Suppose S is the smallest σ -algebra on $\mathbb{R}$ containing $\{(r, r+1) : r \in \mathbb{Q}\}$. Prove that S is the collection of Borel subsets of $\mathbb{R}$.

Proof. By the definition Axler gives for Borel sets, we want to prove that the smallest σ -algebra on $\mathbb{R}$ containing $\mathcal{R} = \{(r, r+1) : r \in \mathbb{Q}\}$ is in fact the smallest σ -algebra containing all open subsets of $\mathbb{R}$. Let $\mathcal{B}$ be the collection of Borel sets. Since the elements of $\mathcal{R}$ are open, certainly $\mathcal{R} \subset \mathcal{B}$.

On the other hand, consider the fact that finite intersections of elements of $\mathcal{R}$ are also in $\mathcal{R}$ to see that $(p, q) \in \mathcal{R}$ where $p, q \in \mathbb{Q}$ such that $q-p < 1$, since $(p, q) = (p, p+1) \cap (q-1, q)$. The set of all such “rational intervals”, call it $\mathcal{U}$, is a basis for the topology of $\mathbb{R}$ (the proof of this fact is routine, but something I’m assuming we can brush past for the purposes of this course). The important thing to realize here is that the open sets of $\mathbb{R}$ can be expressed as unions of elements of $\mathcal{U}$, and since $\mathcal{U}$ is countable, the unions of elements of $\mathcal{U}$ can be expressed as countable unions. Thus, every open set is a countable union of elements of $\mathcal{U} \subset \mathcal{R}$, so every open set is in $\mathcal{R}$. This makes $\mathcal{R}$ a σ -algebra containing all open sets, so $\mathcal{B}$, which is the *smallest* σ -algebra containing all open sets, is a subset of $\mathcal{R}$. $\square$

5. (2.B.11) Suppose $\mathcal{T}$ is a σ -algebra on a set Y and $X \in \mathcal{T}$. Let $S = \{E \in \mathcal{T} : E \subset X\}$.

(a) Show that $S = \{F \cap X : F \in \mathcal{T}\}$.

(b) Show that S is a σ -algebra on X .

Proof. (a) Let $C = \{F \cap X : F \in \mathcal{T}\}$. If $F \in \mathcal{T}$, then $F \cap X \in \mathcal{T}$ since σ -algebras are closed under finite intersections, and thus $C \subset S$. On the other hand, if $E \in S$, then $E \in \mathcal{T}$ and $E \subset X$, so $E = E \cap X$, and we conclude that $S \subset C$.

(b) Let $E \in \mathcal{T}$, $E \subset X$. See that

$$X \setminus E = X \cap (Y \setminus E),$$

and since both X and $Y \setminus E$ are in $\mathcal{T}$, $X \setminus E \in \mathcal{T}$. Also, $X \setminus E \subset X$, so $X \setminus E \in S$, making S closed under complementation in X .

Let $\{E_i\}_{i \in \mathbb{Z}^+} \subset \mathcal{T}$ such that for each i , $E_i \subset X$. Then $\bigcup_{i=1}^{\infty} E_i \in \mathcal{T}$ since $\mathcal{T}$ is closed under countable unions, and $\bigcup_{i=1}^{\infty} E_i \subset X$, so $\bigcup_{i=1}^{\infty} E_i \in S$. Thus, S is a σ -algebra on X . $\square$

6. (2.B.13) Suppose (X, S) is a measurable space, $E_1, \dots, E_n$ are disjoint subsets of X , and $c_1, \dots, c_n$ are distinct nonzero real numbers. Prove that $c_1\chi_{E_1} + \dots + c_n\chi_{E_n}$ is an S -measurable function if and only if $E_1, \dots, E_n \in S$.

Proof. First, take the case $n = 1$. We know from the text (Example 2.32) that χ_{E_1} is S -measurable if and only if $E_1 \in S$. Furthermore, the function $g : \mathbb{R} \rightarrow \mathbb{R}$ defined by $g(x) = c_1x$ is Borel measurable since it is increasing and defined on a Borel set, even when restricted to $\{0, 1\}$ (Theorem 2.43). Since $c_1\chi_{E_1}$ is the composition of a Borel measurable function with an S -measurable function (Theorem 2.44), it is S -measurable.

Proceeding by induction on n , assume that for some n , the theorem holds for $n-1$. Then $c_n\chi_{E_n}$ is S -measurable if and only if $E_n \in S$ by the argument in the first paragraph, and $c_1\chi_{E_1} + \dots + c_{n-1}\chi_{E_{n-1}}$ is S -measurable by hypothesis, then $c_1\chi_{E_1} + \dots + c_{n-1}\chi_{E_{n-1}} + c_n\chi_{E_n}$ is S -measurable (Theorem 2.46). $\square$

Do my current office hours allow you the opportunity to come and ask questions?
Yes!

Name 2 suggested problems where you would like to have a good hint.
2B.7 and 2B.24